Predicting SNAP Payment Errors: A Pilot Proposal for Pennsylvania

Synthetic-data-validated case prioritization for quality-control review

September 29, 2026 · Research whitepaper · Methods and validation artifacts are open-source at github.com/ctrimm/snap-qc-per-experiments

1. Executive summary

We have built and validated a synthetic data generator that reproduces how SNAP payment errors occur in Pennsylvania's caseload, calibrated against five years of federal quality-control (QC) microdata. Machine-learning models trained *only* on this synthetic data predict real Pennsylvania errors meaningfully better than baseline — proof the synthetic cases carry genuine signal about where errors happen.

The operational finding: a two-part ranking model (probability of error × predicted error dollars), applied to prioritize QC review, recovers **$5,483 vs $3,851 for random review** at a fixed review budget — **$67 vs $47 per hour of reviewer time**. Reviewing the riskiest half of cases catches 67% of all errors with half the reviews.

We propose a low-risk pilot: train the ranker on Pennsylvania's own historical QC data, validate on a recent holdout, then run it in shadow alongside existing review. No production eligibility-system changes. No new data collection.

2. The problem

SNAP payment error rates remain stubbornly high nationwide, and every error is either taxpayer money over-issued or a household underpaid. Quality-control review is the corrective mechanism — but reviewer time is finite, and cases are currently selected for review without a data-driven sense of where errors concentrate. The result is expensive review effort spread thinly over a caseload where risk is highly uneven.

Pennsylvania is a strong candidate for a targeted approach. Our analysis of 3,268 Pennsylvania QC cases (FY2020–2024) shows the state's benefit-calculation engine matches federal statute essentially exactly (99.7–100% of cases) — Pennsylvania's errors live in the *human process layer* (verification,

timing, documentation), which is precisely where a prediction tool adds value. Error patterns are also concentrated: shelter-deduction errors run at roughly twice the national rate, while computation errors run at a tenth.

3. What was built and validated

A Pennsylvania-conditioned synthetic generator. Starting from a nationally calibrated generator (error-type mix matching reality within ~1 percentage point), we fit Pennsylvania-specific error-rate offsets and empirical error mixes. The PA-conditioned output reproduces Pennsylvania's finding mix, element mix, and cause mix within a few percentage points, while leaving national output unchanged.

Validated predictive signal. Models trained only on PA-conditioned synthetic cases predict real, held-out Pennsylvania errors: 0.556 element-level accuracy vs a 0.453 baseline, and PA-conditioning genuinely beats a nationally trained model. The one honest threat to this result — whether the generator had indirectly seen the holdout data — was tested with a fully clean refit; the advantage survived.

A dollar-aware ranking. The set's most important surprise: ranking cases by error probability *alone* performs worse than random at the top slice. Error probability and error size are nearly orthogonal in Pennsylvania's mix — the dollars live in income errors, not the frequent shelter errors. The product must therefore be a two-part ranker: calibrated probability of error multiplied by predicted error dollars.

Headline pilot economics (simulated on held-out PA cases, top-20% review budget):

	Random review	Two-part ranker
Error dollars recovered	$3,851	**$5,483**
Value per review hour	$47	**$67**
Share of all error dollars captured	—	28.4%

Of the $5,483: most is over-issuance (direct program savings); the remainder is under-issuance (households underpaid — the correct QC outcome, but a make-whole cost). PA5's as-run split was $4,239 over-issuance / $1,835 under-issuance (~70/30), measured before the

> generator update described in the revision note below; it has not been recomputed under current defaults.
>
> **Revision note (October 3, 2026).** The headline economics above were revised after exp44's fitted household-size mix was adopted into the generator (*src/synth/household.py*) after PA5 had run: PA5's published $6,074 recovered / $75 per review hour / 31.5% of error dollars used the legacy mix. Re-running PA5's exact two-part ranker recipe on the same holdout under current generator defaults (verified in exp47) gives $5,483 recovered, $67/hr, 28.4% of error dollars — a 1.42x lift over random review's $3,851 / $47/hr, vs PA5's 1.58x. The lift direction and the qualitative economics are unchanged. PA5's experiment chronicle is preserved unchanged as the as-run record; this whitepaper reflects the system as it exists today.

4. The pilot

What Pennsylvania provides

- Historical QC case extracts the state already holds (no new data collection, no new PII).
- A review team willing to work a ranked queue for a defined pilot period.

What changes — and what does not

- **Phase 0 (cold start, no PA data needed):** the synthetic-trained ranker scores a sample of cases; rankings are validated against known outcomes.
- **Phase 1:** the two-part ranker is trained on PA's historical QC data and validated on a recent holdout the model never saw.
- **Phase 2 (shadow):** the ranker scores the live review caseload; reviewers work the top slice; outcomes are measured against the baseline selection method.
- **What does not change:** no modifications to the eligibility system, no automated determinations (the model prioritizes review — it never decides a case), no data leaves the state's control.

Success metrics

- Precision in the reviewed slice vs baseline selection.
- Error dollars recovered per reviewer hour vs baseline.

- Reviewer time saved to reach a fixed error-capture target.

A reasonable pilot shape is 8–12 weeks from data handoff to shadow-readout.

5. Limitations and honest caveats

- **Headroom.** The synthetic-trained ranker captures roughly one-third of the performance a model trained on real PA data is projected to reach. The pilot's Phase 1 exists to close exactly this gap.
- **Small-dollar skew.** The errors the ranker catches skew toward smaller dollar amounts; large-dollar errors remain harder to find and are the priority for Phase 1 feature work.
- **Procedural errors are invisible.** Published federal error rates run 2–3 points above what case microdata implies, entirely in procedural errors the microdata cannot see. This tool targets substantive case errors, not procedural compliance.
- **Stability.** The signal is year-to-year stable in our tests, but the error-probability baseline should be re-tuned on recent data at pilot start.
- **Human judgment stays in the loop.** This is a triage tool for reviewers, not an adjudication engine. Final determinations remain human — by design.

Household-type review burden (exp47, October 2026)

A dollar-maximizing ranker is dollar-efficient and risk-regressive at the same time. Stratifying the holdout by household composition (elderly, disabled non-elderly, single-parent, married with kids, multi-adult with kids, childless adult), elderly and disabled households have the highest error rates (PA: 0.48 and 0.54; NY: 0.34 and 0.42) but the lowest review rates relative to their risk: review-to-risk ratios of **0.22 and 0.33 in PA** — the elderly ratio worse than random allocation (0.42) — and **0.34 and 0.35 in NY**. Childless adults, with the lowest error rates, are reviewed at 1.41x (PA) and 1.45x (NY) their risk.

The mechanism is the dollar objective itself, not a ranking bug: elderly/disabled errors are frequent but small-dollar (medical deductions, timing), while dollars concentrate in income errors — so maximizing dollars per review hour rationally deprioritizes the highest-risk groups. Within each type the ranker is precise (0.62–0.73 precision on reviewed elderly/disabled cases in PA) and selects cases carrying 2–14x the predicted dollars of unselected cases of the same type. The pilot implication: if the goal includes finding underpayments owed to vulnerable households — elderly and disabled households carry high under-issuance exposure — the dollar ranker works against that

goal while optimizing the headline $/hr. A type-aware review budget or a blended dollars-plus-equity objective is the honest next step for Phase 2 design.

6. The ask

We are looking for a Pennsylvania partner to run this pilot: a data contact who can provide historical QC extracts, and a review team for the shadow phase. We bring the validated methodology, the synthetic cold-start model, and the open-source validation artifacts. All methods, experiments, and results are published openly; nothing about the approach is proprietary or black-box.

Methods, experiment chronicles (50+), and validation code: github.com/ctrimm/snap-qc-per-experiments (private research repo; access on request). This work is conducted in the open-source civic-technology lane. Contact: via the repository.